RSET

RAJAGIRI SCHOOL OF ENGINEERING & TECHNOLOGY (AUTONOMOUS)

B. TECH. DEGREE PROGRAMME
IN
(Common to)
CS, IT

SEMESTER VI

(2023 ADMISSIONS)

SYLLABUS

Rajagiri Valley, Kakkanad,
Kochi 682 039, Kerala, INDIA
www.rajagiritech.ac.in

COURSE CODE	COURSE NAME	L	T	P	SL	CREDIT	YEAR OF INTRODUCTION
102902/ CO600D	AI Concepts	3	0	0	4.5	3	**2023**

1. Preamble

This course introduces students to the fundamentals of knowledge representation, problem solving, and learning methods in artificial intelligence. This course also equips students with the foundational concepts and techniques used in natural language processing. Students will gain an understanding of generative models and their applications in AI.

2. Prerequisite

Nil

3. Syllabus

Module 1: Introduction to AI (8 Hours)
Overview of AI, Problems, Problem space, and searching techniques, Definition of production system, Control strategies, Heuristic search techniques, Game Playing-Minmax search procedure, Alpha-beta cutoff, Intelligent agents: structure and functions Agents and environment, Types of agents: Simple reflex agent, Goal-based agent, Utility-based agent, Learning agents, Knowledge-Based Agent.

Module 2: Knowledge Representation (9 Hours)
Knowledge representation issues, Representation and mapping, Approaches to knowledge representation, Using predicate logic, Representing simple facts in logic, Representing instance and ISA relationships, Computable functions and predicates, Resolution, Natural deduction. Representing knowledge using rules, Procedural vs. declarative knowledge, Logic programming, Forward vs. backward reasoning, matching, control knowledge**.**

Module 3: Probabilistic Reasoning and Expert Systems (7 Hours)
Representing knowledge in an uncertain domain, Semantics of Bayesian networks, Dempster-Shafer theory, Planning overview, Components of a planning system, Classical Planning - Goal stack planning, Hierarchical planning, Conditional Planning, Probabilistic Planning, Continuous or Temporal Planning.

Expert Systems: Representing and using domain knowledge, Expert system shells, Knowledge acquisition.

Module 4: Natural Language Processing (8 Hours)
Introduction to NLP, Fundamentals of NLP, Feature representation in NLP, Sequential Models for NLP, Transformers: Fundamentals and Attention Mechanisms, Transformers: Architecture and Variants.

Module 5: Generative AI (8 Hours)
Introduction to Generative AI, Generative AI & Pretrained Models, Fine-Tuning Techniques, Prompt Engineering, Introduction to Large Language Models (LLMs)- From Transformers to LLMs, BERT architecture, Ethical issues in AI, Explainable AI(XAI).

Module 6: Applications (5 Hours)
Agentic AI, Responsible AI, Introduction to LangChain, AI prompts using zero-shot, few-shot, and chain-of-thought prompting, AI Tools – CrewAI and AutoGen. Retrieval-Augmented Generation (RAG), Frameworks: PyTorch, TensorFlow, Hugging Face, OpenAI API.

4. Textbooks

1. Elaine Rich, Kevin Knight, Shivashankar and B. Nair, *Artificial Intelligence*, Tata McGraw Hill, 3rd Edition, 2017.
2. Stuart J. Russell and Peter Norvig, *Artificial Intelligence: A Modern Approach,* Pearson Education Asia, 4th Edition, 2022.
3. Daniel Jurafsky, James H. Martin, *Speech and Language Processing: An Introduction to Natural Language Processing, Computational Linguistics, and Speech Recognition with Language Models*, 3e, 2025.
4. Altaf Rehmani, *Generative AI for Everyone: Understanding the Essentials and Applications of This Breakthrough Technology*, 2024.

5. Reference Books

1. Dan W. Patterson, *Introduction to Artificial Intelligence & Expert Systems*, 13th edition, Pearson Education, 2023.
2. George F. Luger, *Artificial Intelligence: Structures and Strategies for Complex Problem Solving*, 6th edition, Pearson Education, 2021.
3. Numa Dhamani, *Introduction to Generative AI,* , Kindle Edition, 2024.
4. Raymond Lee, *Natural Language Processing: ATextbook with Python Implementation*, Second Edition, Springer, 2025.

5. Allen, James, *Natural Language Understanding*, Benjamin/Cummings, 2nd edition, 1995.
6. Akmajian, A., R. A. Demers, A. K. Farmer, and R. M. Harnish, *Linguistics – An Introduction to Language and Communication,* 7e, The MIT Press, 2017.

6. Course Outcomes

After the completion of the course the student will be able to

CO1: Apply appropriate search strategies to solve AI application problems and explain the functioning of various types of intelligent agents within different environments.

CO2: Apply various knowledge representation techniques, including predicate logic and rule-based systems and deduce solutions using the principle of resolution.

CO3: Apply probabilistic reasoning and planning techniques and analyze the concepts of expert systems.

CO4: Understand and apply fundamental concepts of Natural Language Processing.

CO5: Understand the fundamentals of Generative AI.

CO6: Understand different AI prompting methods and tools to build useful AI applications.

7. Mapping of Course Outcomes with Program Outcomes

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	3	2	2							2
CO2	3	3	2	2							2
CO3	3	3	1	1							2
CO4	2	2	1								2
CO5	2	2	1	1	2	1	2				2
CO6	2	2			3		1	2	2		2

8. Mapping of Course Outcomes with Program Specific Outcomes

	PSO1	PSO2	PSO3
CO1			2
CO2			2
CO3			2
CO4		1	2
CO5		2	2
CO6		1	1

9. Assessment Pattern

Learning Objectives	Continuous Internal Evaluation (CIE)		End Semester Examination (ESE out of 100)
	Internal Examination 1 (50)	**Internal Examination 2 (50)**	
Remember	5	5	10
Understand	20	35	60
Apply	25	10	30
Analyse			
Evaluate			
Create			

10. Mark Distribution

Total	CIE					ESE
	Attendance	**Internal Examination**	**Assignment /Quiz**	**Project/ Report from Module 6**	**Total**	
150	10	25	10	5	50	100

11. End Semester Examination Pattern

There will be two parts; Part A and Part B. Part A contains 10 questions with 2 questions from each module, having 3 marks for each question. Students should answer all questions. Part B contains 2 questions from each module of which student should answer any one. Each question can have a maximum of 2 sub-divisions and carry 14 marks.

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